

Class 6

Raster Data Analysis

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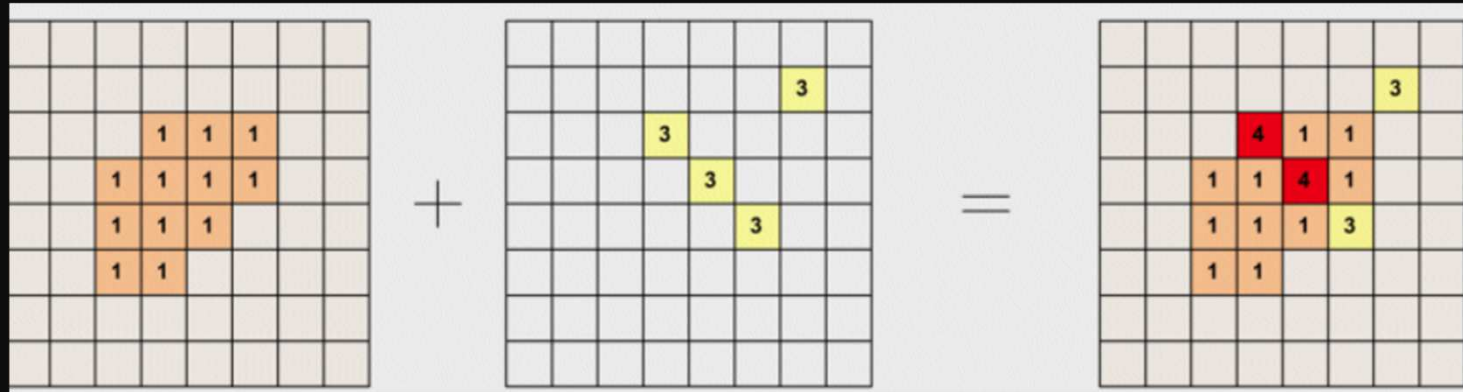
Lab for Spatial Informatics, IIIT, Hyderabad

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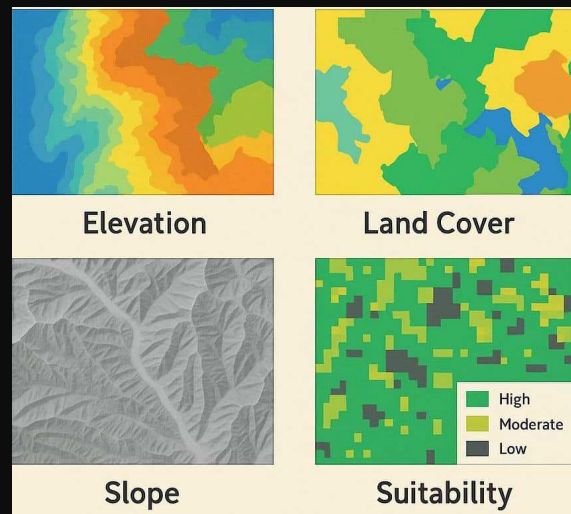
Raster Analysis

Raster GIS operations have several strengths:

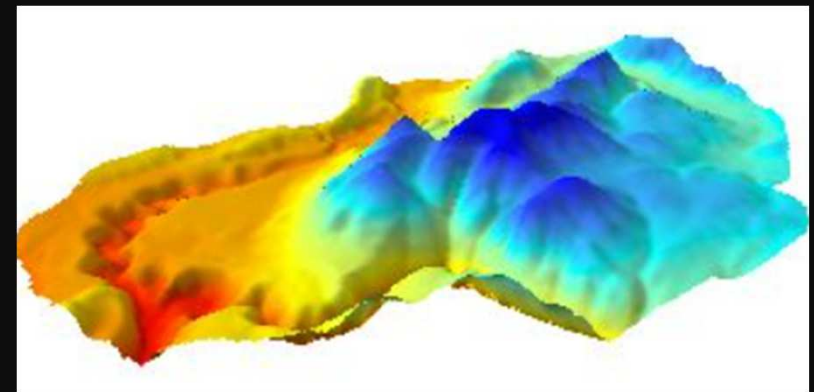
1. Efficient for overlaying different themes
2. Allows mathematical operations using cell values from multiple layers



3. provides a frame for modeling dispersion across space and for surface modeling

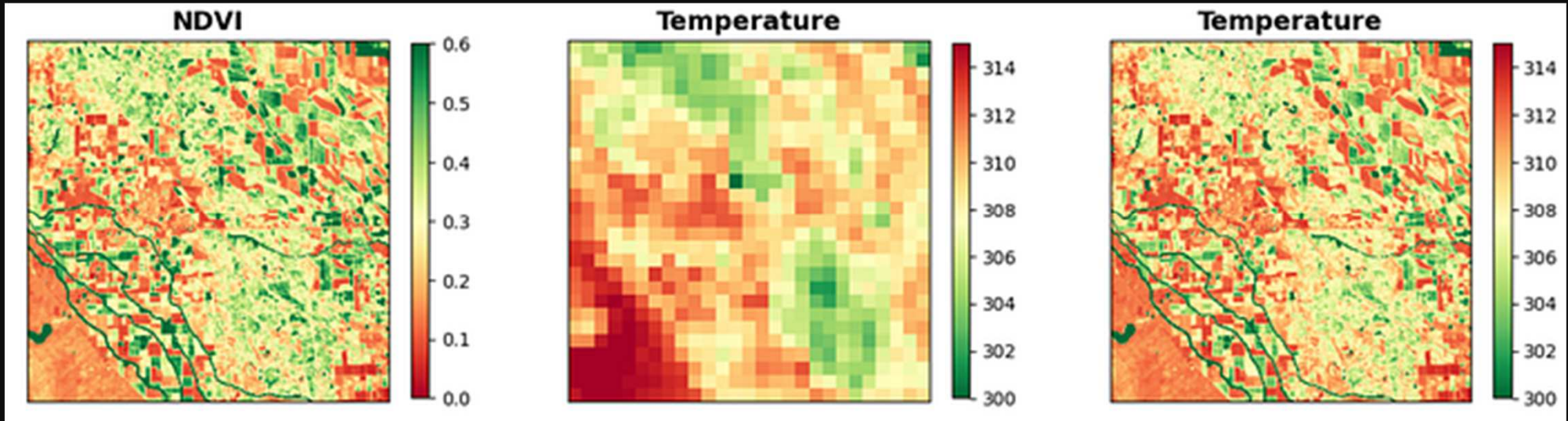


4. enables 3D visualization



Raster Functions/Operations

- Each Raster Layer represents a thematic information
 - sometimes, the associated properties/ attributes



Sentinel-2 NDVI (10 m)

Sentinel-3 temperature (1000 m)

downscaled Sentinel-3
temperature (10 m)

Map Algebra (Mathematical operations
across multiple raster layers)

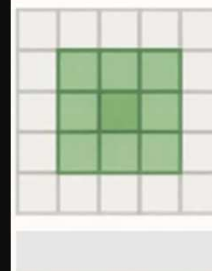
Neighbourhood Function;

Euclidean;

Metric (distance);

Topological

Raster Neighborhood Analysis



- **Focal Sum/Mean**
Computes sum
or average

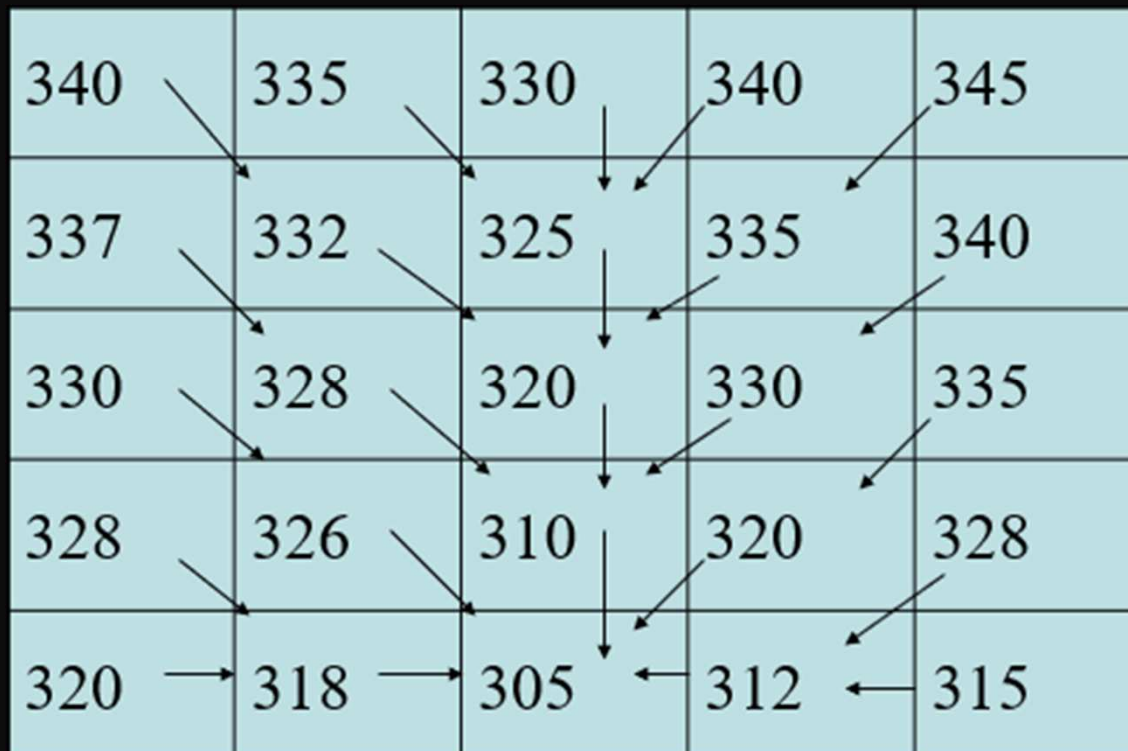
Conducted using a moving window

- **Focal Statistics**
Computes statistics for each cell
using its neighbors

- **Filter**
Smooths or
enhances using
low pass filter

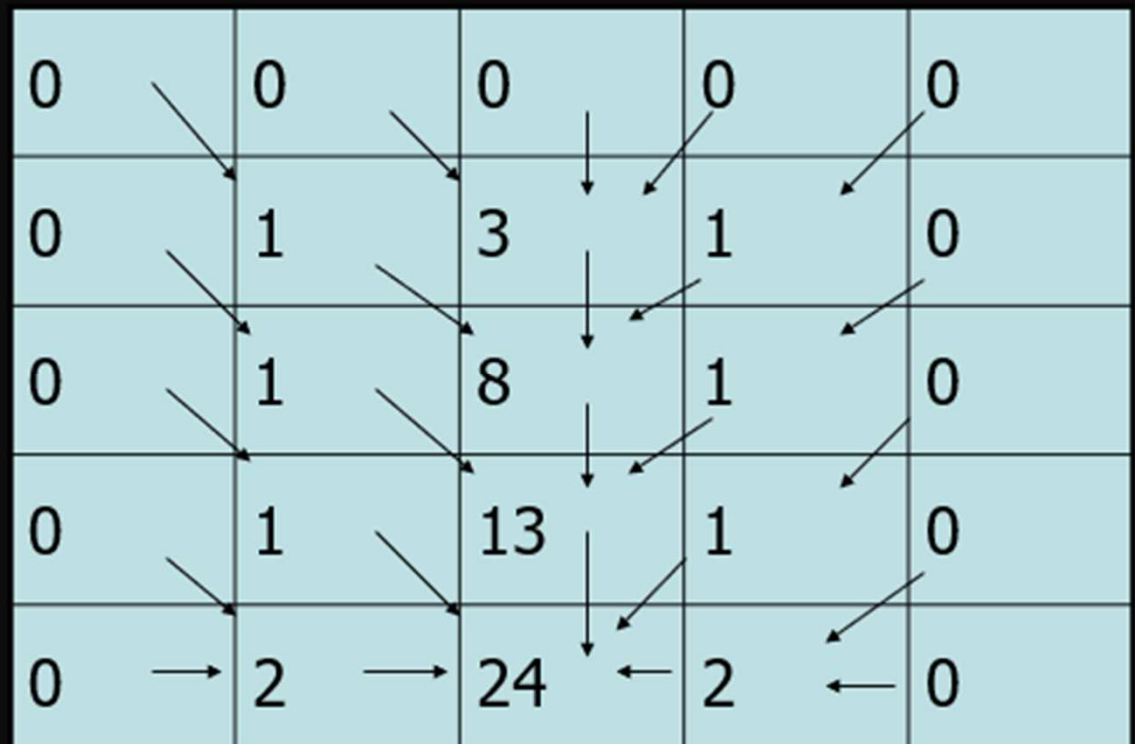
- **Focal Sum/
Mean**
Computes sum
or average

6	=	=	=
6	6	=	6
Filter		Focal Flow	
→	↓	13	13
→	6	13	11
Focal Sum/mean		Focal /Mean	



Flow Direction

Flow accumulation



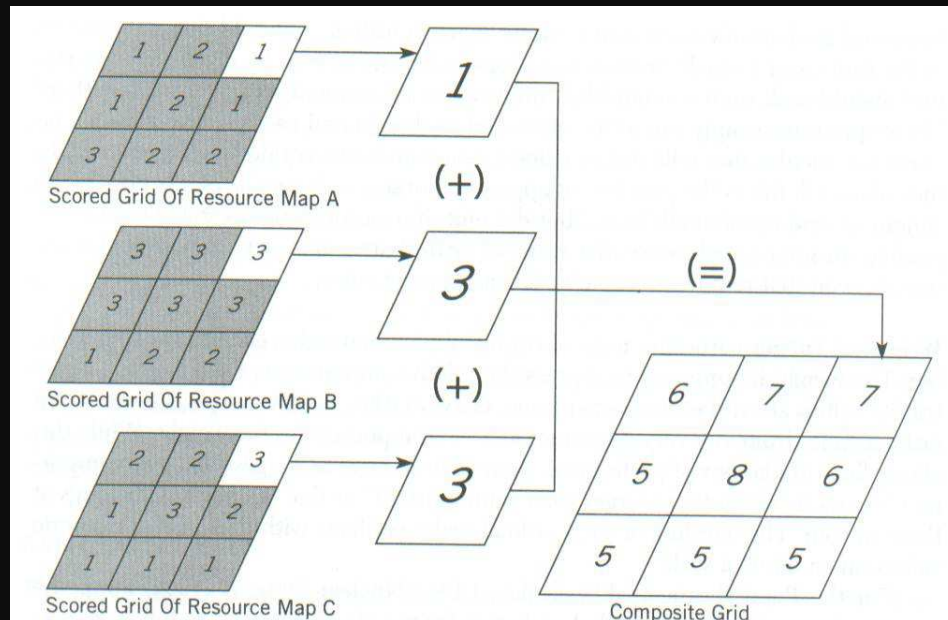
Raster GIS: Map Algebra Operators and Functions

MAP ALGEBRA OPERATORS:

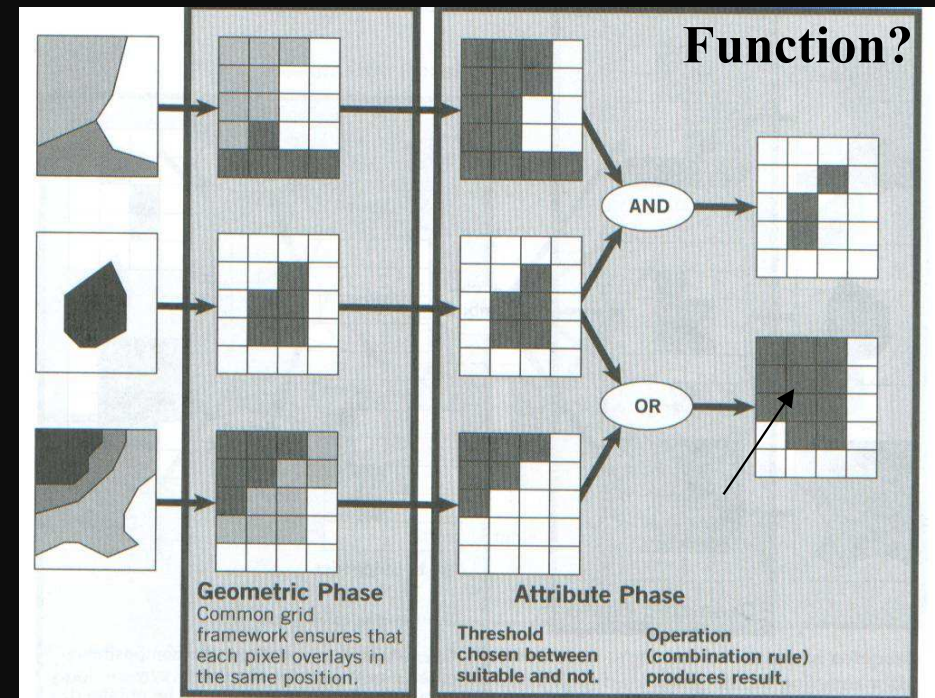
arithmetic	-	$+$, $-$, $*$, $/$, $^$
boolean	-	True or False
comparison	-	$<$, $<=$, $=$, $>$, $>=$
logical	-	contained within sets
combinatorial	-	overlay of different grids with attributes
accumulative	-	sum, divide, multiply groups of values
assignment	-	assignment of values during operations

Raster Based Overlay

Simple Addition



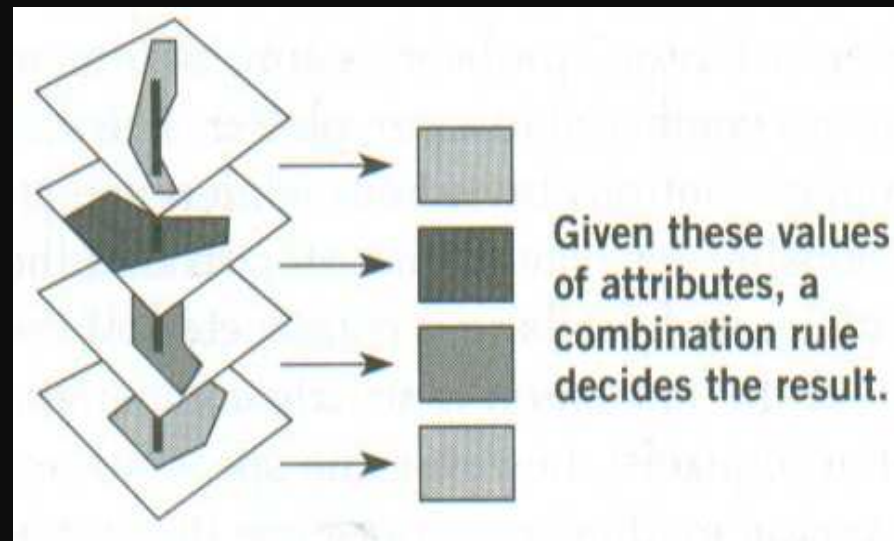
Boolean Combine

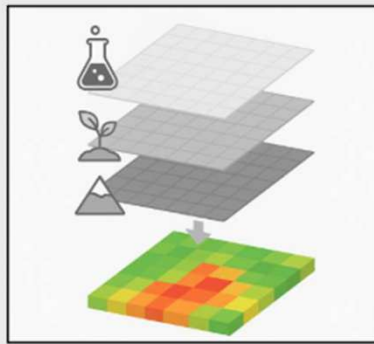


Cell by cell evaluation of mathematical functions

Overlay: Combining Attributes

Select attributes of interest for a given location



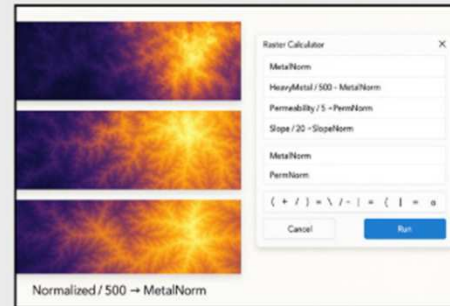


Suppose you need to create an Environmental Contamination Risk Index. You start with 3 raster datasets:

- 🧪 Heavy Metal Concentration (0-500 mg/kg)
- 🌱 Soil Permeability Rating (1-5)
- 🏔️ Terrain Slope (0-20%)

Each one holds crucial environmental data for every cell on the map.

Visualizing the Layers



Each raster layer shows different physical or chemical properties of the land.

🛠️ In your GIS interface:

- Grayscale maps represent values
- Legends indicate scale (e.g., dark = high value)

These rasters help us understand pollution, water flow, and terrain risk.

Normalize Your Data



HEAVYMETAL / 500

0-1



PERMEABILITY / 5

0-1



SLOPE / 20

0-1

Apply a Weighted Formula

$$\text{RISK INDEX} = 0.4 \cdot \text{HMC} + 0.3 \cdot \text{PERM} + 0.3 \cdot \text{SLOPE}$$



0.4 * HEAVY METAL
CONCENTRATION

+



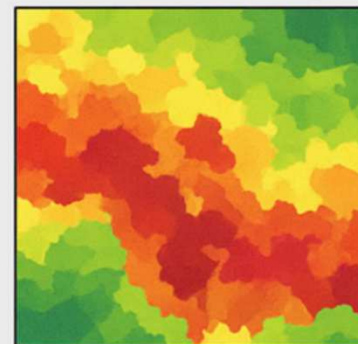
0.3 * PERMEABILITY

+



0.3 * SLOPE

Final Risk Map

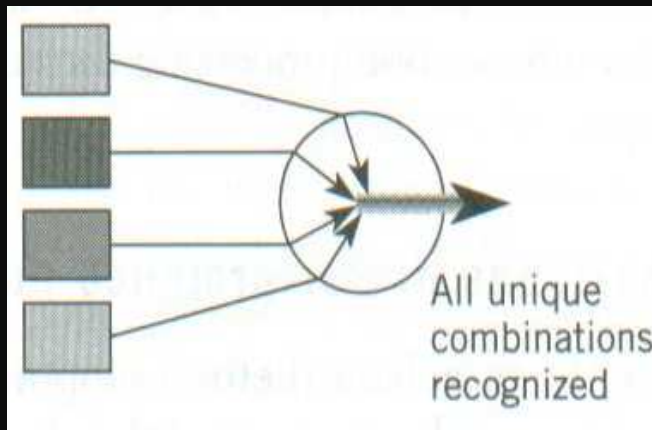


📍 It combines all three inputs into a single map showing contamination hotspots:

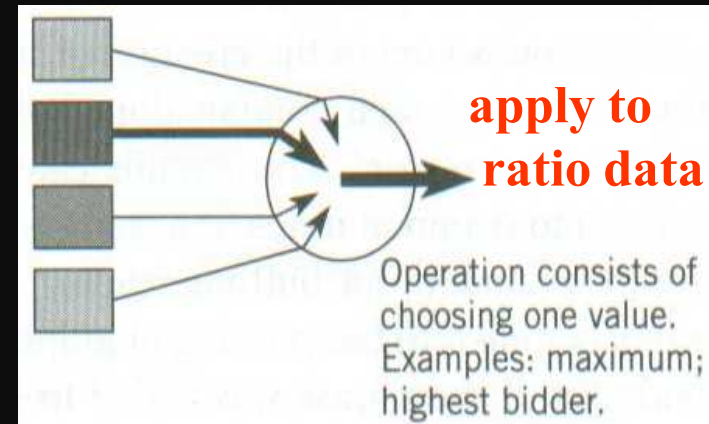
- 🔴 Red zones = high risk
- 🟡 Yellow = moderate
- 🟢 Green = low risk

Overlay: 4 Basic Rules

#1 Enumeration Rule: Each Attribute preserved in output



#2 Dominance Rule: One value wins



5.3	6.2	6.7	8.1
1.1	1.4	5.6	6.6
6.5	7.4	8.2	9.1
3.3	5.5	7.7	6.2

Factor 1

and

6.1	7.5	6.2	7.1
3.1	2.4	7.6	5.6
6.3	7.5	8.0	5.1
2.3	6.5	5.7	5.2

Factor 2

6.1	7.5	6.7	8.1
3.1	2.4	7.6	6.6
6.5	7.5	8.2	9.1
3.3	6.5	7.7	6.2

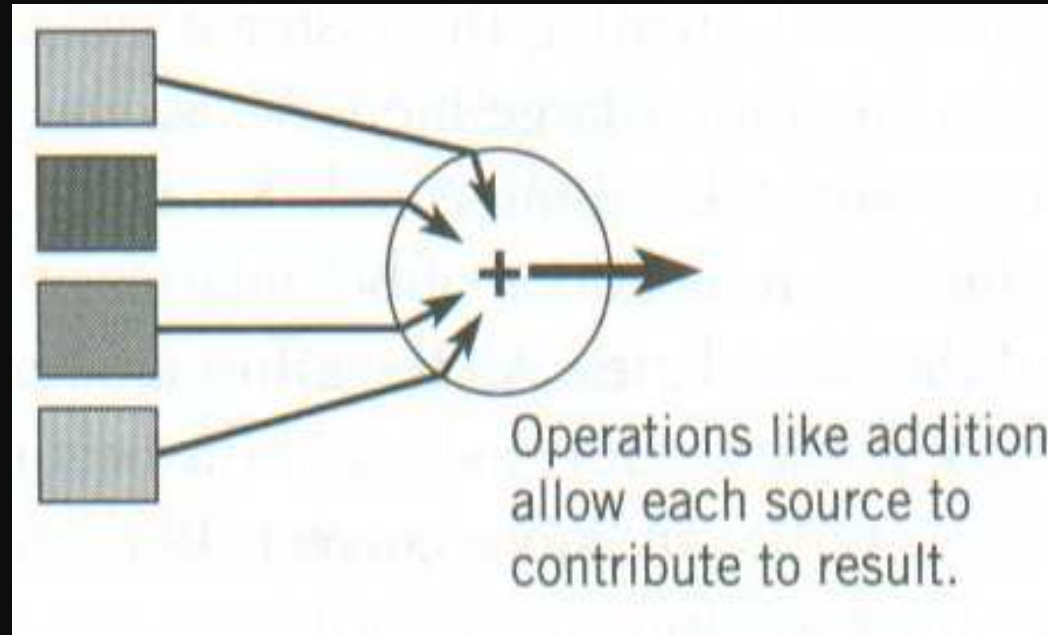
highest
bid

2	2	1	1
2	2	2	1
1	2	1	1
1	2	1	1

highest
bidder

Overlay: 4 Basic Rules

#3 Contributory Rule: each attribute value contributes to result



nominal categories

1	1	0	0
0	0	1	0
0	1	1	1
1	1	1	0

+

0	1	0	0
0	1	0	1
0	1	0	1
0	0	1	0

=>

1	2	0	0
0	1	1	1
0	2	1	2
1	1	2	0

Landslide susceptibility mapping using frequency ratio method in Sikkim Himalaya

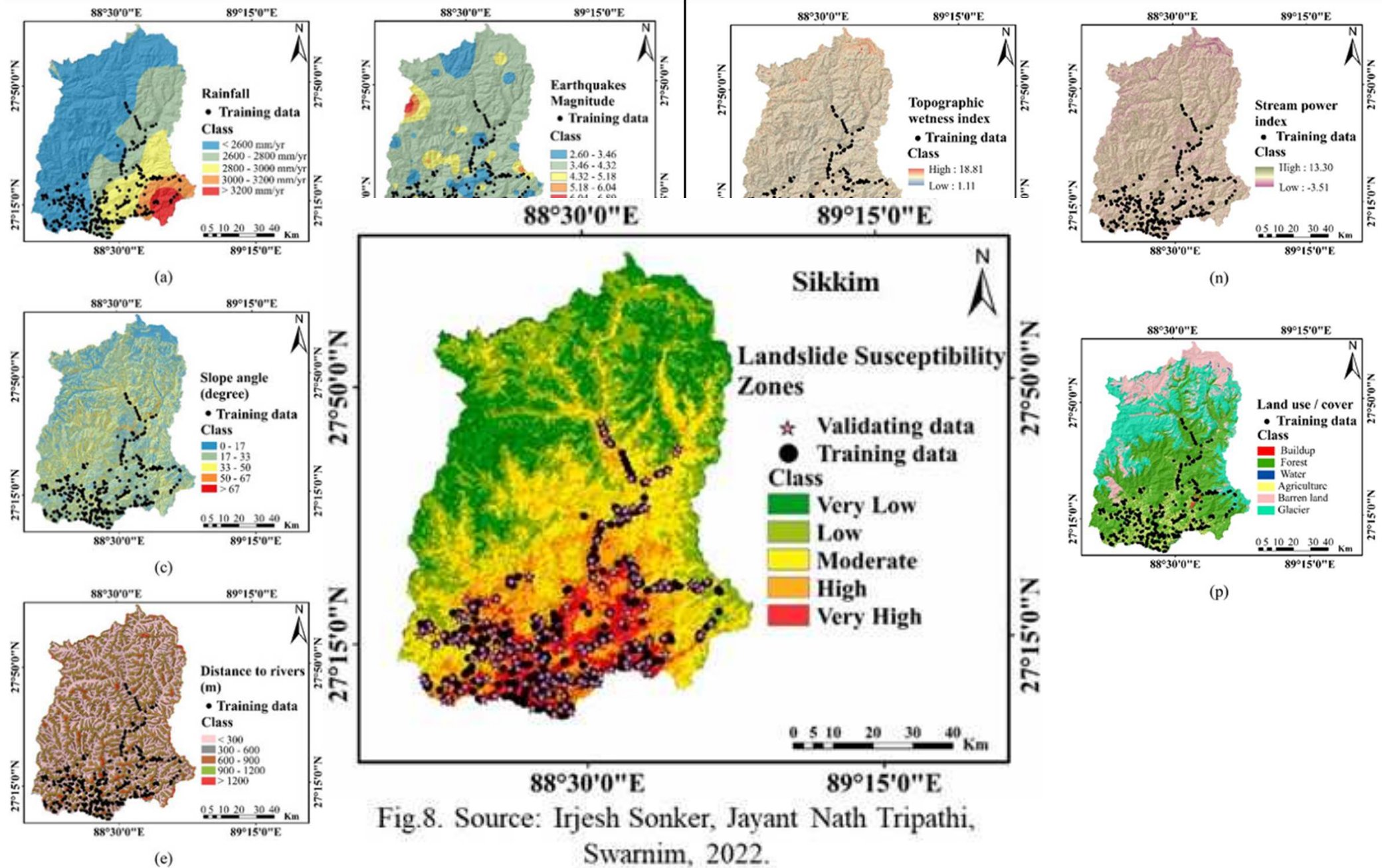


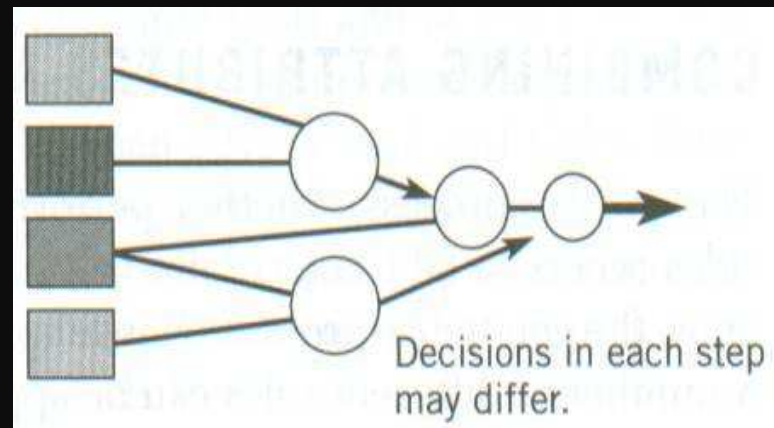
Fig.8. Source: Irjesh Sonker, Jayant Nath Tripathi, Swarnim, 2022.

<https://doi.org/10.1016/j.qsa.2022.100067>

#4 Interaction Rule: pair of values contribute to result

Factor combination

all possible combinations are considered and rated



1	2	1	2
3	3	1	3
1	3	1	2
1	1	1	2

and

3	1	1	1
2	1	3	3
1	1	2	1
1	1	1	3

$\Rightarrow$

3	4	1	4
8	7	3	9
1	7	2	4
1	1	1	6

legend

1	:	1	&	1
2	:	1	&	2
3	:	1	&	3
4	:	2	&	1
5	:	2	&	2
6	:	2	&	3
7	:	3	&	1
8	:	3	&	2
9	:	3	&	3

Cell/Pixel/Grid based operations

- Local Operations
- Focal Operations
 - Depends on attributes at x & $f(x)$ and the attributes in the neighbourhood $n(x)$
 - Results in a focal operation $\phi(x)$
- Zonal Operations
 - Aggregates the value over a set of zones

Global (Incremental)

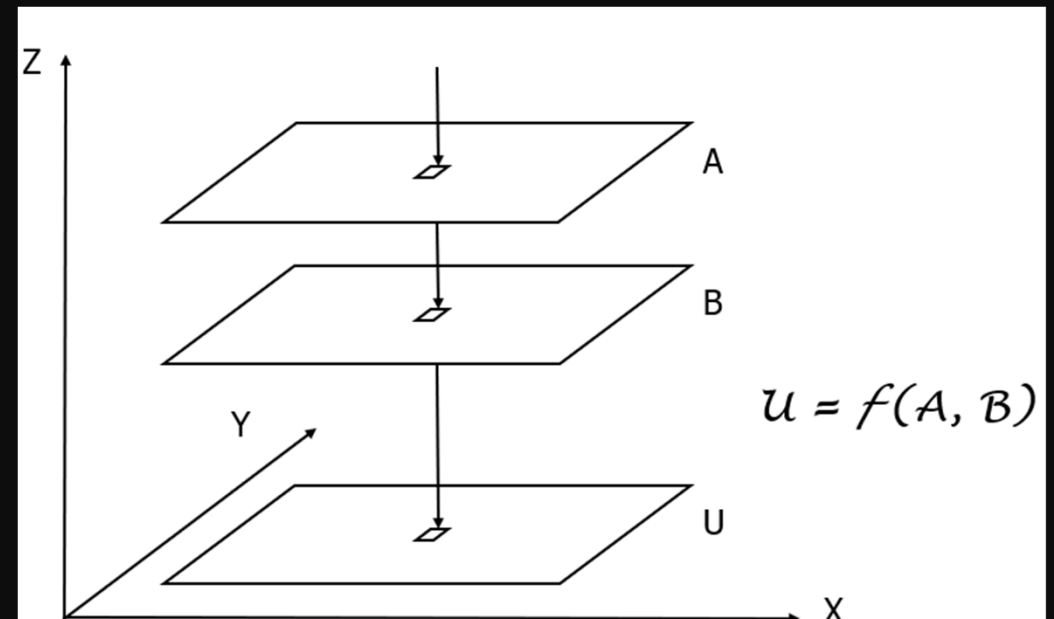
Local functions

- Local functions operate on the values of all the attributes relating to each cell (location).
- The operations are independent of the effects of attribute values from neighbouring cells.
- A local function results in a new grid as a function of one or more input grids.

The generic form of local functions

$$U = f(X1, X2, \dots)$$

Spatial context of local functions



Ex: `new_map = old_map_1 + old_map_2`

Local functions

Single Raster and Multiple Raster (Map Algebra)

Summary statistics including max, min, range, sum, mean, median and SD – Raster with numeric data;

Majority, minority for raster with numeric and categorical data

Local Functions: Example

•Multiply by constant value

2	0	1	1
2	3	0	4
	1	1	2
		3	2

X 3 =

6	0	3	3
6	9	0	12
	3	3	6
		9	6

•Multiply by a grid

2	0	1	1
2	3	0	4
	1	1	2
		3	2

X

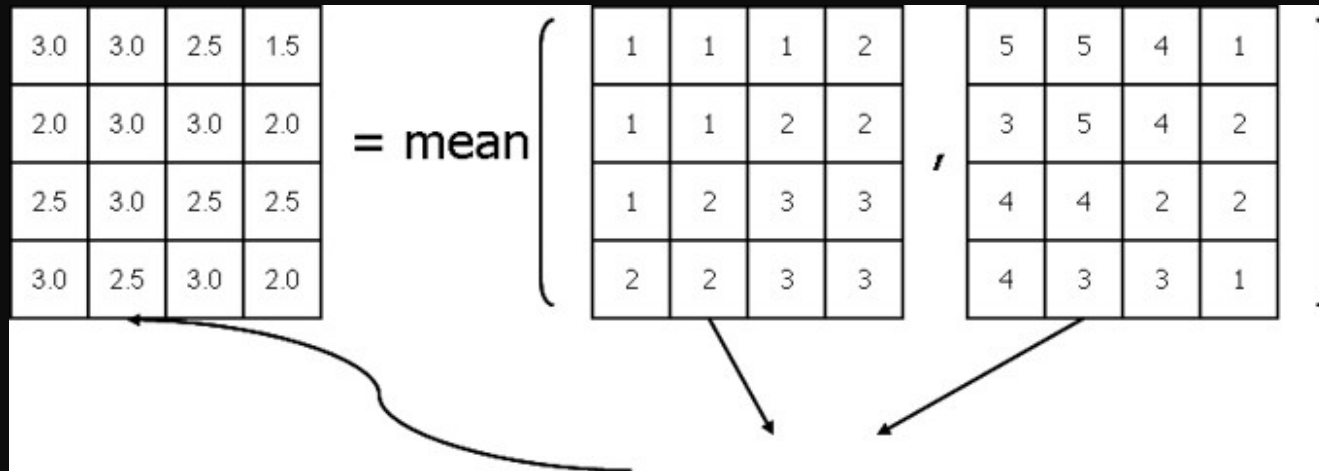
2	0	2	2
3	3	3	3
	2	2	2
		1	1

=

4	0	2	2
6	9	0	12
	2	2	4
		3	2

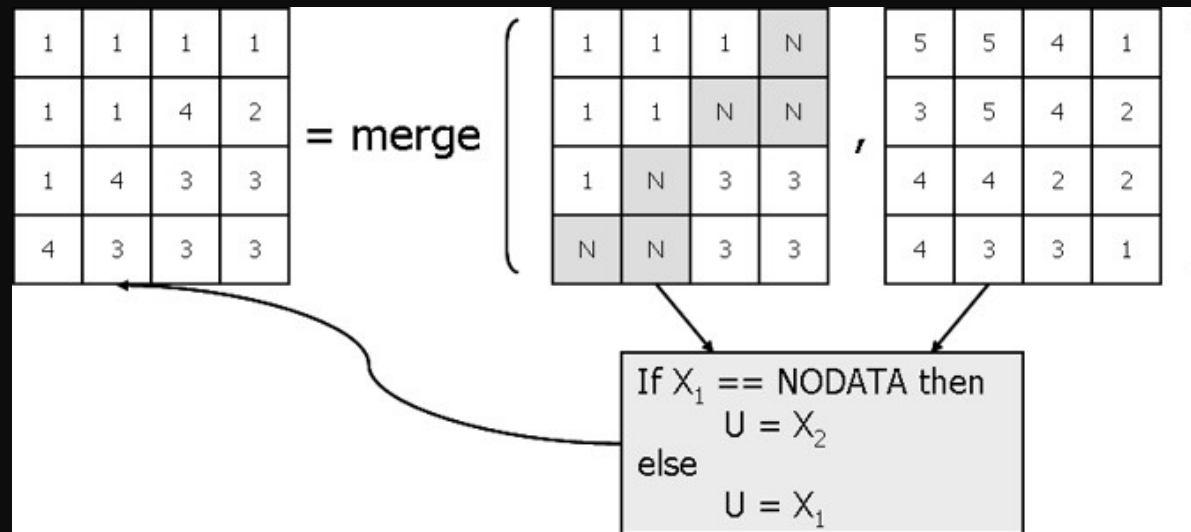
Local operations that do not involve statistics or computation use operation called “Combine” Which assigns a unique output value to each unique combinations of input values (Ex. Slope and aspect)

Average



Eg. `outgrid = mean(ingrid_1, 2 * ingrid_2, (ingrid_3 + ingrid_4))`

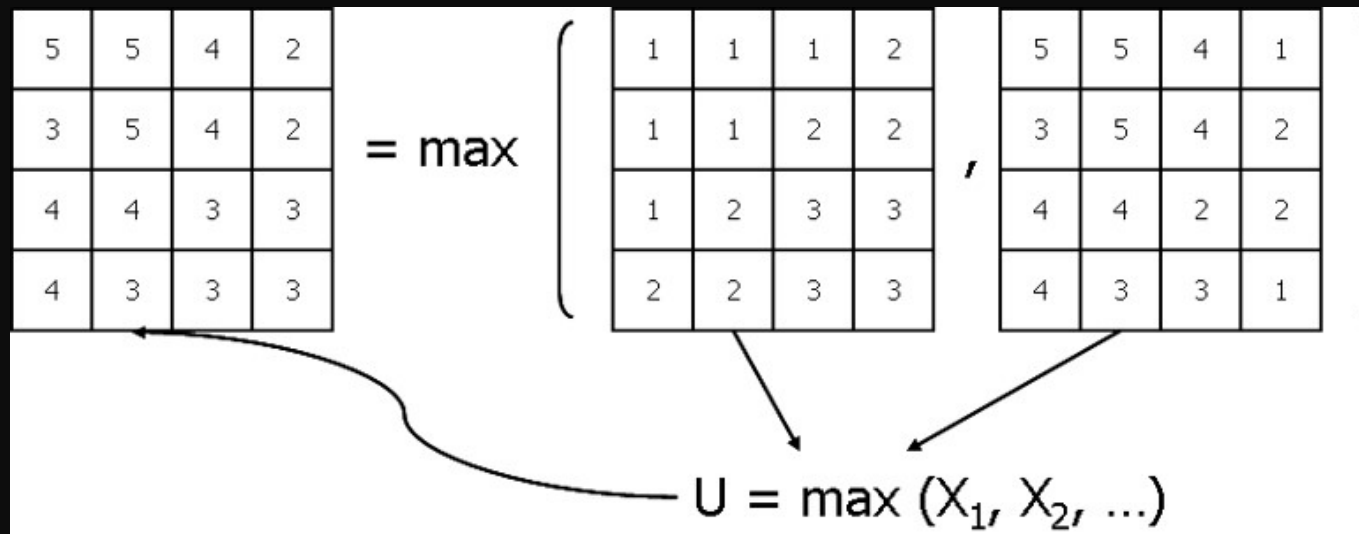
Merge



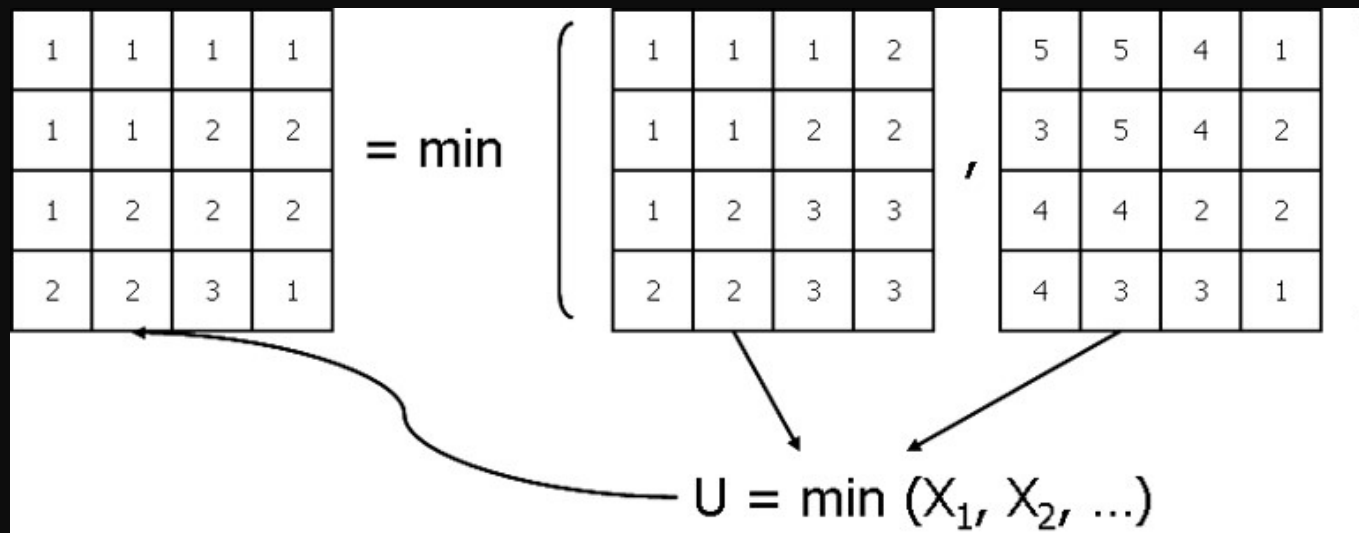
Eg. `outgrid = merge(ingrid_1, ingrid_2, ...)`

Maximising

Eg. $\text{outgrid} = \max(\text{ingrid_1}, \text{ingrid_2})$

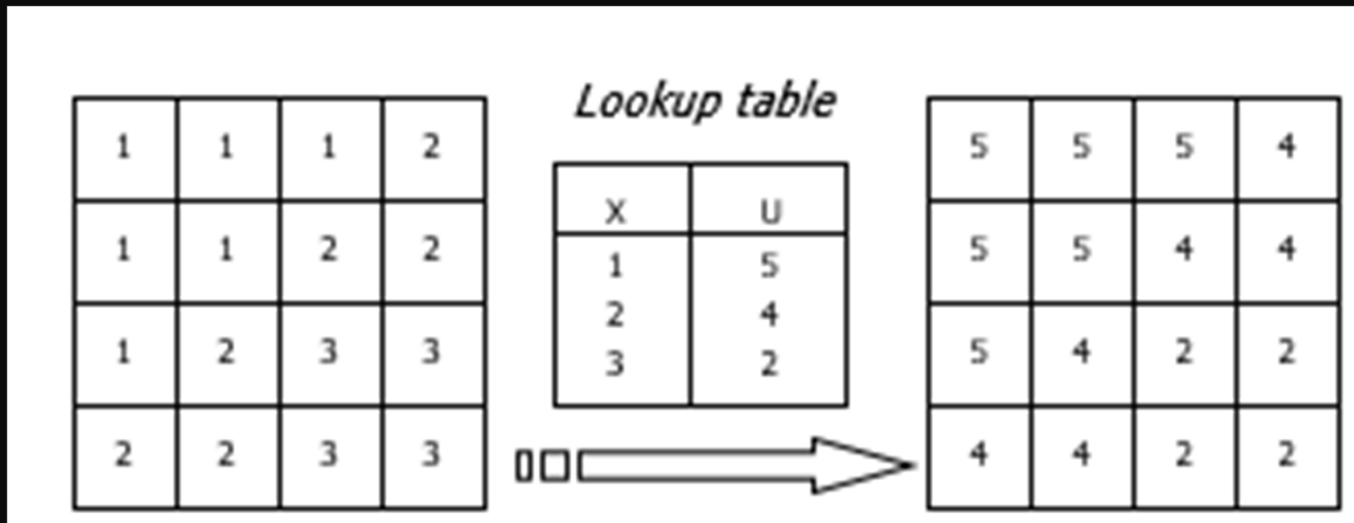


Minimising



Eg. $\text{outgrid} = \min(\text{ingrid_1}, \text{ingrid_2})$

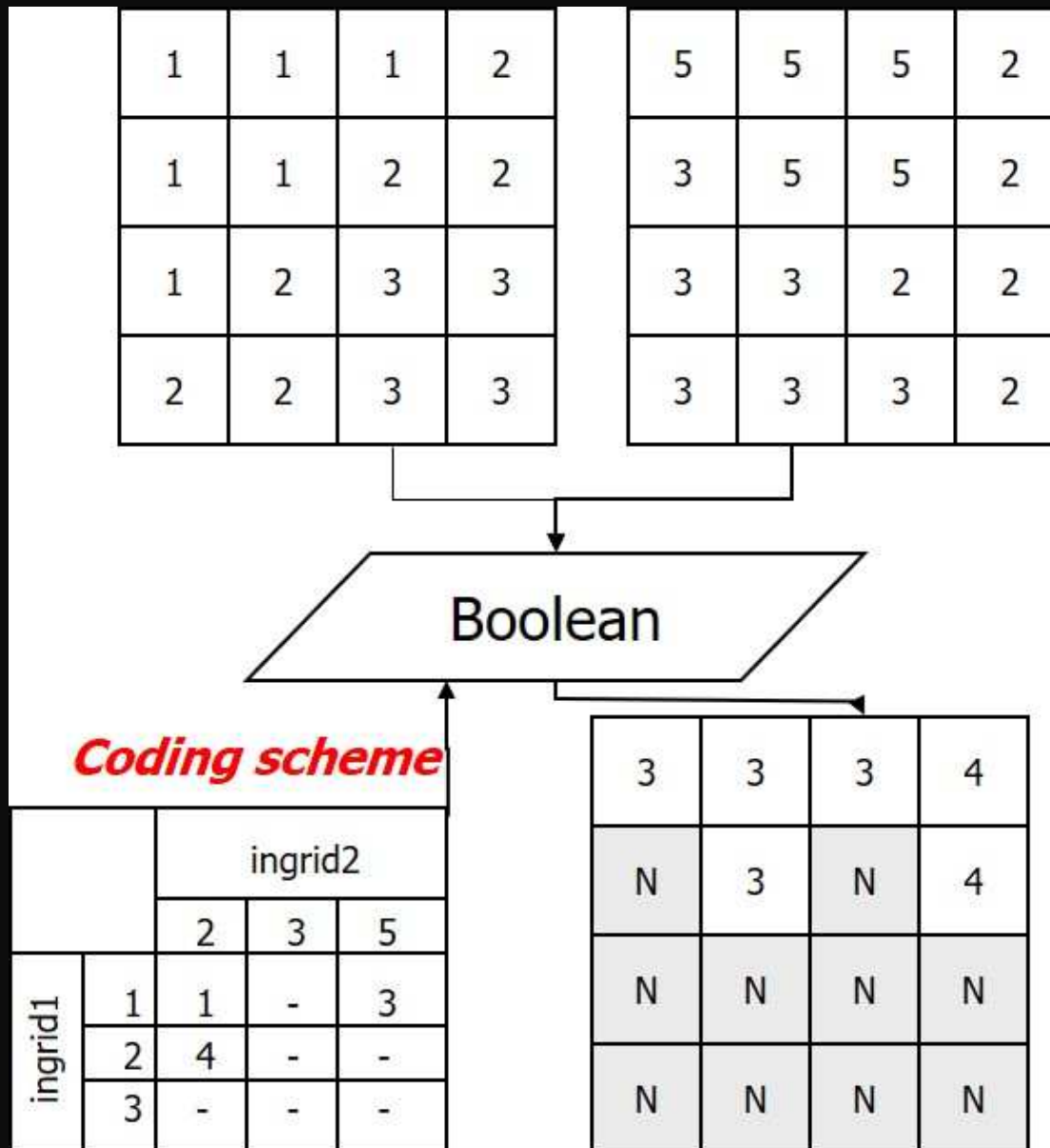
Reclassification – arbitrary



Given $a_1, a_2, \dots, a_n$
 and $b_1, b_2, \dots, b_n$
 if $X == a_1$ then
 $U = b_1$
 else if $X == a_2$ then
 $U = b_2$
 ...
 else
 $U = X$

Eg. If (ingrid == 1)
 outgrid = 5
 else if (ingrid == 2)
 outgrid = 4
 else if (ingrid == 3)
 outgrid = 2
 endif
 or outgrid = con(ingrid == 1, 5, ~
 con(ingrid == 2, 4, 2))

Reclassification – Boolean



Given $a_1, a_2, \dots, a_n; b_1, b_2, \dots, b_n;$

$c_1, c_2, \dots, c_n$

if $X_1 == a_1$ and $X_2 == b_1$ then

$U = c_1$

else if $X_1 == a_2$ and $X_2 == b_2$ then

$U = c_2$

...

else

$U = \text{NODATA}$

Eg. If ($\text{ingrid1} == 1$ & $\text{ingrid2} == 2$)

$\text{outgrid} = 1$

else if ($\text{ingrid1} == 2$ & $\text{ingrid2} == 2$)

$\text{outgrid} = 4$

else if ($\text{ingrid1} == 1$ & $\text{ingrid2} == 5$)

$\text{outgrid} = 3$

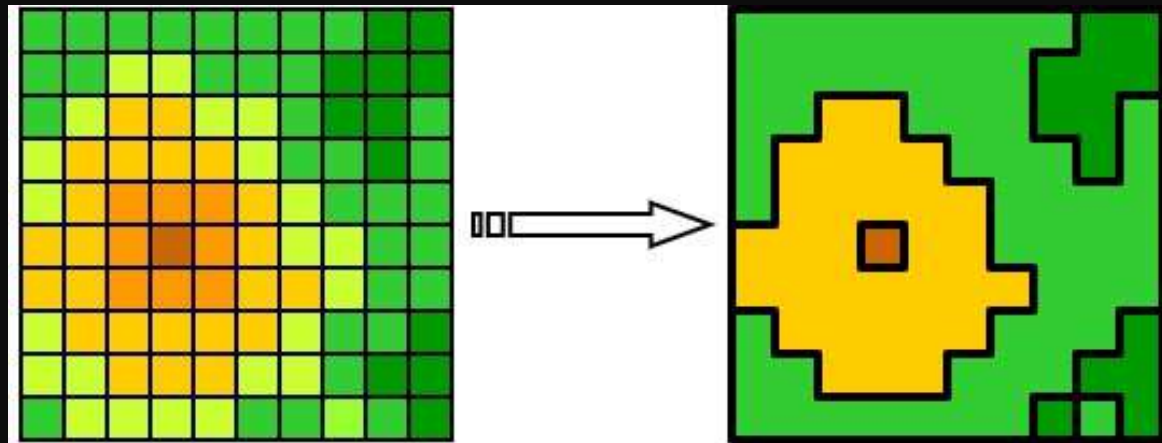
endif

Reclassification – statistical

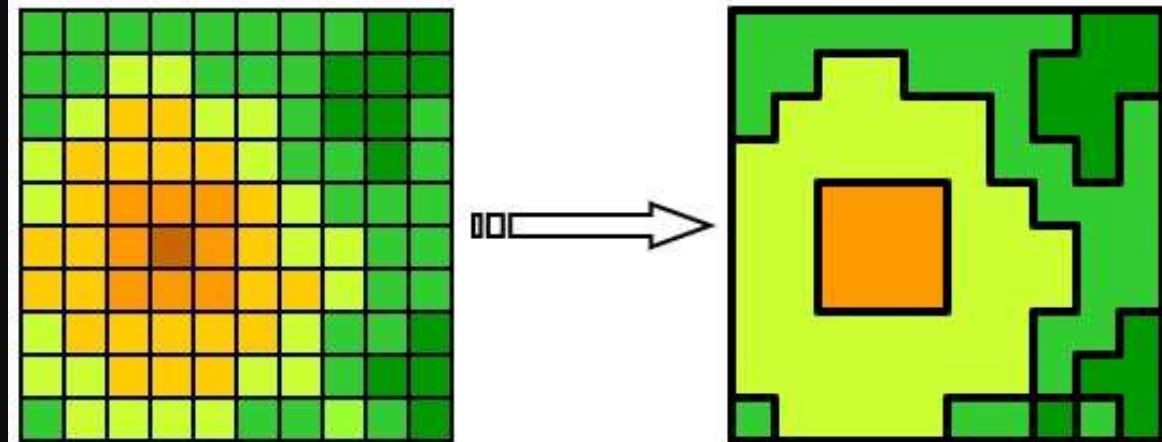
Slice

Divide range values
in either equal intervals
or equal areas

```
Outgrid = slice(ingrid, ~  
EQAREA | EQINTERVAL, ~  
nzones, base_zone#, ~  
in_min, in_max)
```



Slice in equal intervals



Slice in equal areas

Advantages of Reclassification

1. Creates simplified raster

Ex. Slope values

2. Creates raster that has unique category values

Ex. Irrigated crop land = 1

3. Creates new raster that shows ranking of cell values in input raster

Ex. 1 to 5 with 1 being least and 5 being most suitable

Reclassifying slopes

